

Lesson 9-3: Write Equations to Represent Relationships Between Two Variables

Grade 6 Mathematics · Standard 6.AT.11

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

Recreation World charges \$24.95 per month. What is the total cost of 2 months of membership?

- ☐ A \$24.95
- ☐ B \$26.95
- ☐ C \$49.90
- ☐ D \$74.85

Question 2 SELECTED RESPONSE · 1 POINT

In the equation $c = 24.95m$ for the gym membership, what does m represent?

- ☐ A The number of months of membership
- ☐ B The total cost in dollars
- ☐ C The monthly fee of \$24.95
- ☐ D The name of the gym

Question 3 SELECTED RESPONSE · 1 POINT

A canoe rents for \$15 per hour. Let h = hours and c = total cost. Which equation represents the relationship?

- ☐ A $c = 15h$
- ☐ B $c = 15 + h$
- ☐ C $h = 15c$
- ☐ D $c = h - 15$

Question 4 SELECTED RESPONSE · 1 POINT

Using $c = 15h$, what is the total cost of a 3-hour canoe rental?

- ☐ A \$15
- ☐ B \$18
- ☐ C \$45
- ☐ D \$75

Question 5 SELECTED RESPONSE · 1 POINT

In the gym membership situation, which quantity is the DEPENDENT variable?

- ☐ A The number of months
- ☐ B The total cost
- ☐ C The monthly fee of \$24.95
- ☐ D The number of gyms in town

Question 6 SELECTED RESPONSE · 1 POINT

The membership table shows: 1 month, \$24.95; 2 months, \$49.90; 3 months, \$74.85. By how much does the total cost increase each month?

- ☐ A \$24.05
- ☐ B \$24.95
- ☐ C \$25.00
- ☐ D \$49.90

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

A spin studio charges \$7.50 for each class. Let n be the number of classes a member takes and let c be the total cost in dollars. Which equation represents the relationship?

Fill in the bubble next to the one best answer.

- ☐ **A** $c = 7.5 + n$
- ☐ **B** $n = 7.5c$
- ☐ **C** $c = n \div 7.5$
- ☐ **D** $c = 7.5n$

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which reasoning justifies the equation chosen in Part A?

- ☐ **A** The number of classes is added to the price of one class, so 4 classes cost $7.5 + 4 = \$11.50$.
- ☐ **B** The number of classes depends on the cost, so the classes belong alone on the left side of the equation.
- ☐ **C** The \$7.50 is shared among the classes, so 4 classes cost $7.5 \div 4 = \$1.875$ each.
- ☐ **D** The \$7.50 is charged for every single class, so the total is 7.5 times the classes; 4 classes cost $7.5 \times 4 = \$30$.

Question 8**SELECT ALL THAT APPLY**

A community pool charges \$9 for each hour of lane rental. Let h be the number of hours rented and let c be the total cost in dollars. Select ALL statements that correctly represent this relationship.

Mark the box next to EVERY answer that is true. More than one is correct.

- ☐ **A** $c = 9h$
- ☐ **B** $h = 9c$
- ☐ **C** The total cost is 9 times the number of hours rented.
- ☐ **D** $c = 9 + h$
- ☐ **E** In the equation, h stands for the number of hours rented and c stands for the total cost in dollars.
- ☐ **F** In the equation, c stands for the \$9 charged for each hour.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.